

SECTION-D

Note: Long answer type questions. Attempt any one questions out of two questions. (1x10=10)

Q.19 Find the equation of circle passing through (0,0), (1,3) and (4,-2) (CO4)

Q.20 Find the middle term in the expansion of $(x + \frac{1}{x})^{12}$ (CO3)

No. of Printed Pages : 4

188412

Roll No.

Level 3, 1st Sem / DVOC (Ref. & Air Cond., Auto. Servicing. ITM, PT, SD, AMT, FP, EMS)

Subject : Applied Mathematics - I

Time : 2 Hrs.

M.M. : 50

SECTION-A

Note: Multiple Choice questions. All questions are compulsory. (5x1=5)

Q.1 If $A = \{1,3,5,7\}$ and $B = \{2,3,4\}$, then find $A \cap B$. (CO1)

- a) $\{3\}$ b) $\{1,3,5\}$
- c) $\{2,3\}$ d) None of these

Q.2 The missing term in the AP 2,5,8,_____, 14 is_____ (CO2)

- a) 12 b) 10
- c) 11 d) 13

Q.3 $i^6 =$ _____. (CO3)

- a) 1 b) -1
- c) i d) $-i$

Q.4 The slope of straight line passing through two points (x_1, x_2) and (y_1, y_2) is _____ (CO4)

- a) 0 b) 1
c) $\frac{y_2 - y_1}{x_2 - x_1}$ d) $\frac{y_2 + y_1}{x_2 + x_1}$

Q.5 If M is maximum value and m be the minimum value then the range of data is _____ (CO5)

- a) $m + n$ b) $m - n$
c) Both of these d) None of these

SECTION-B

Note: Objective type questions. All questions are compulsory. (5x1=5)

Q.6 The value of $\tan 45^\circ$ is _____. (CO1)

Q.7 $2+4+6+8+\dots$ is AP. (True/False) (CO2)

Q.8 $5! - 3!$ _____. (CO3)

Q.9 The slope of line which makes an angle of 45° with positive x-axis is _____. (CO4)

Q.10 If a coin is tossed two times then the sample space is _____. (CO5)

SECTION-C

Note: Short answer type questions. Attempt any six questions out of Eight questions. (6x5=30)

Q.11 In a triangle ABC if $BC = 1$ cm, $AC = 2$ cm and $AB = 3$ cm then find cosine of angle A. (CO1)

Q.12 If the 2nd and 5th term of GP are 9 and 243 then find the series. (CO2)

Q.13 Prove that $2^n > n$ for positive integer n by using Principal of mathematical induction. (CO3)

Q.14 Find equation of straight line whose intercepts are $a = 9$ and $b = 3$. (CO4)

Q.15 Two unbiased coins are tossed simultaneously find probability of getting one tail. (CO5)

Q.16 Define isosceles and equilateral triangle with the help of diagram. (CO1)

Q.17 Use Binomial theorem to find the value of $(101)^4$. (CO3)

Q.18 Find the sum of 10 terms of the series $6+9+12+15+\dots$ (CO2)